



FSKM 2025

ISP568 FUZZY LOGIC SYSTEMS

**Faculty of Computer Science and Mathematics****Group Project**

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COVID-19 PREVENTION DOOR SYSTEM		

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Item Assessed	Weight	Score (1-4)
1. <u>Introduction</u>		
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Significance (Entrepreneur Value / Marketable Solution)	1	
2. <u>Literature Review</u>	1	
3. <u>Methodology</u>		
Knowledge Acquisition	1	
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4. <u>Result and Analysis</u>	2	
5. <u>Conclusion</u>	1	
6. <u>References</u>	1	
TOTAL SCORE	25	/100



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ISP568 FUZZY LOGIC SYSTEMS



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FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

**CDCS259 BACHELOR OF INFORMATION SYSTEM (HONS.)
INTELLIGENT SYSTEMS ENGINEERING**

ISP568 FUZZY LOGIC SYSTEMS

GROUP PROJECT: COVID-19 PREVENTION DOOR SYSTEM

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1.0 INTRODUCTION

1.1 PROJECT BACKGROUND

Health and safety are becoming more preferred as areas in public and private settings after the COVID-19 pandemic. Not all manual screening procedures such as taking the temperature using a handheld scanner, or asking people to wear a mask, are always reliable and can lead to close contact that increases infection risk. In order to solve the problem a contactless and automatic system is required to assist in limiting access to a building. Our project presents a smart door system that utilises fuzzy logic in making decisions on two broad categories which is body temperature and mask usage. Fuzzy logic facilitates managing cases that cannot be safely or unsafe categories such as when one's temperature is high or even when the mask is not worn properly. This system checks both values using sensors and opens a door only if the person meets safety conditions. This concept could be applied if the person not only stops further spread of COVID-19 but for the future health emergencies as well.

1.2 PROBLEM STATEMENT

For the time of Covid-19 pandemic, it was proved that it was necessary to provide access to shared indoor facilities without the presence of physical contact. Health experts say that social distancing, checking for high temperature, and insisting on use of masks are basic steps towards limiting the spread of the virus (WHO, 2020). Furthermore, configuring these requirements manually requires a lot of work, more time, and increased risk of frontline workers getting infected with the virus.

One of the best priorities is to work out a system capable of controlling access to entry health criteria by using smart and automated procedures. Traditional systems that function at preset levels are limited by lack of flexibility and inherent imprecision, and to a large extent this is due to factors such as environmental influence or involuntary errors by individuals. The need to automate schemes to track health related data effectively is required with a high level of accuracy and the uptake of multiple health inputs for informed decision making. Now, by applying fuzzy logic, the system can handle uncertainties and variation in people's actions and



from their sensors' readings, hence ensuring that only people who meet health and safety requirements gain access (Khokhar et al., 2023).

1.3 OBJECTIVES

To support Covid-19 prevention efforts, this project aims to design and implement a fuzzy logic-based automatic door system that enhances health screening and access control. The specific objectives are as follows :

1. To determine the fuzzy value and variable for the Covid 19 Prevention Door System
2. To design fuzzy graphs and fuzzy rules for Covid 19 prevention door system using Matlab
3. To apply a fuzzy expert system for Covid 19 prevention door system using the fuzzy Mamdani method.

1.4 SIGNIFICANCE (ENTREPRENEUR VALUE / MARKETABLE SOLUTION)

The proposed automated door system that uses fuzzy logic solves a critical need in the current and post-pandemic world by providing a contactless, intelligent health screening that also can enhance safety in public and private indoor environments. This system's ability to evaluate both the use of masks and body temperature with fuzzy logic decision-making provides a unique market that leads to advantage by combining accuracy with ease of use. This technology makes it possible to attract for use in the offices, hospitals, shopping complex, and any open crowded area.

From an entrepreneurial point of view, this solution may bring to the growing demand market for smart health and safety technologies, which has seen growing faster due to the COVID-19 pandemic. Key elements that contribute to its marketability are:

1. Enhanced Safety and Security

- Advanced Health Screening: Contactless health screening reduces physical contact and virus spread. Automated door operation only allows access to the individuals that meet safety criteria, to prevent unsafe entry.



- Boosting confidence: Improving the safety system may boost people's trust in the security of interior spaces, and at the same time encouraging a return to normal activities and assisting in economic recovery.

2. Public Health Promotion

- Market demand: this system encourages a highly marketable and scalable. It can be used in various settings, from small offices to large open public areas. And its applicability beyond COVID-19 to future emergencies health, offering long-term potential growth.
- Automating the enforcement on wearing masks and keeping social distancing protocols (system promotes public health): It can reduce human error and negligence in screening processes, to ensure a culture of safety and increasing public confidence in public areas.

3. Cost efficiency

- Uses low cost microcontrollers and sensors, it can reduce implementation costs.
- It can reduce workers costs by automating health checks. It can also help improve the economic recovery.

4. Adaptability

- A flexible fuzzy logic system can make changes to additional health parameters in the future.
- It can be customized according to different environments needs and changing health guidelines.

5. User Centric Design

- The intuitive interface ensures an excellent user experience because of clear instructions and feedback to encourage health and safety norms.
- Customisable Access Options: Based on different needs and preferences of various user groups such as employees in business settings or residents in the residential design, the system can be customized.



2.0 LITERATURE REVIEW

The COVID 19 breakout has radically transformed the global outlook on health and safety practices especially in shared indoor spaces. As identified in the study by the authors of the report “Automated Door to Prevent COVID-19 using Fuzzy Logic” (n.d.), control of entry/exit into buildings with regard to health parameters like temperature and masks is critical in reducing virus spread. Conventional screening techniques such as physical temperature checking and oral mask prompting are not only unreliable but also heighten chances of contact with frontline-workers. In order to overcome this challenge, the development of smart and automated health screening systems is being increasingly explored.

The application of fuzzy logic is one of the main concepts in this project. Formulated for the first time in the year 1965 by Lotfi Zadeh, fuzzy logic has been applied to the fields dealing with uncertainty-free decision-making. Unlike binary logic systems, fuzzy logic also helps to incorporate partial truth values which lie between 0 and 1, for example, useful for applications such as health screening, where sensor readings in terms of temperature or detection of mask may be inaccurate and subjective. In this project, fuzzy logic is used in the computation of the input data, like body temperature and mask status so as to find out if an individual has met the safety level to enter.

The previous work had shown the plausibility to use infrared and image processing for the non-contact health monitoring. For instance, there have been the use of infrared thermometers and thermal cameras in open places such as airports to monitor people with fever, which is among the main symptoms of COVID-19. Similarly the use of machine learning and computer vision techniques have been used to recognize face masks with high accuracy in real time.

The implementation of such sensor technologies into an automatic door system enhances security as well as efficiency. Doors that are automated and capable of opening based on the results of health screenings eliminate the requirement of human contact and promotes contactless access, thus adhering to social distancing norms. Such a system comes in handy especially where



there is a high volume of traffic like in hospitals, offices, and public places where keeping health security is critical.

Fuzzy logic has also been in other health related systems other than Covid-19 screening. For example, in a study by El-Sattar(2021), fuzzy techniques were used to improve face images to improve recognition accuracy especially when under poor lighting conditions or low resolution image setting. This could play a key role in any system that involves visual inputs like face mask detection. The use of fuzzy making allows the systems to remain functional even if the input data is noisy, making it suitable for use in real world environments such as buildings or clinic entrances.

In the article by Duth et al.(2021), a smart door system of contactless temperature detection, mask detection, and automatic sanitization using UV-C light was proposed. Through their work, the possibility of integrating several sensors and microcontrollers to create an end to end pandemics access control system solution was proved.

In addition to temperature and mask detection, in some additional research health indicators have been added such as cough detection and heart rate monitoring. Although these can make the system more complicated, they also improve its accuracy and ability to detect individuals who are potentially at risk but asymptomatic. Accessing additional inputs into a fuzzy logic system can enhance the rule base and improve the door control output, ensuring that only individuals that meet the criteria are allowed to enter.

Other than that, the proposed system has an entrepreneurial and market value. The document mentions the rise of demand for smart health and safety technologies because of the pandemic as the system can be used for further health crises. Using cost-effective microcontrollers such as Arduino and Raspberry Pi and incorporating cheap sensors, the system is affordable and scalable to other environments. Its flexibility, module nature and the contactless design make it a suitable option for public health infrastructure upgrade.



After all, the application of fuzzy logic to automated access systems deals, at once, with questions of health and security in common areas. The strategy offers an alternative approach to well-known screening methods and further promises to have other applications outside the current pandemic.



3.0 METHODOLOGY

3.1 KNOWLEDGE ACQUISITION

Knowledge acquisition for this project started with an extensive survey of existing literature on health screening methods related to COVID-19 prevention, especially non-contact screening procedures such as infrared thermometer and thermal imaging.

Automated prevention door systems include the collection of information from many aspects that are important for design and implementation. This helps establish the importance of automated, contactless systems in order to reduce virus transmission risks. In addition, automated door systems also need to know how to interpret the data from different sensors including temperature sensors, camera (mask detection sensor). This includes understanding different types of sensors that are used to determine the level of feverishness and whether that person is wearing a mask or not.

These inputs then process into a fuzzy logic model whose membership function and fuzzy rules were defined through knowledge from the expert and health guideline to determine whether individuals met the safety requirement for entry or not.

The fuzzy logic system was stimulated in MATLAB in order to validate the accuracy in decision making and then experimentally using microcontroller such as the ATmega328P, temperature sensors, mask detection sensor in order to test the performance of the system in real life situations. This research involves evaluating health safety, compatibility, and cost factors in ensuring that the selected components are aligned with the project's objectives.



3.2 KNOWLEDGE REPRESENTATION

3.2(A) SEMANTIC NETWORK

https://drive.google.com/file/d/1Q8zgU7MV5Duacld_Jf2YcMWtM6GvdAzb/view?usp=sharing

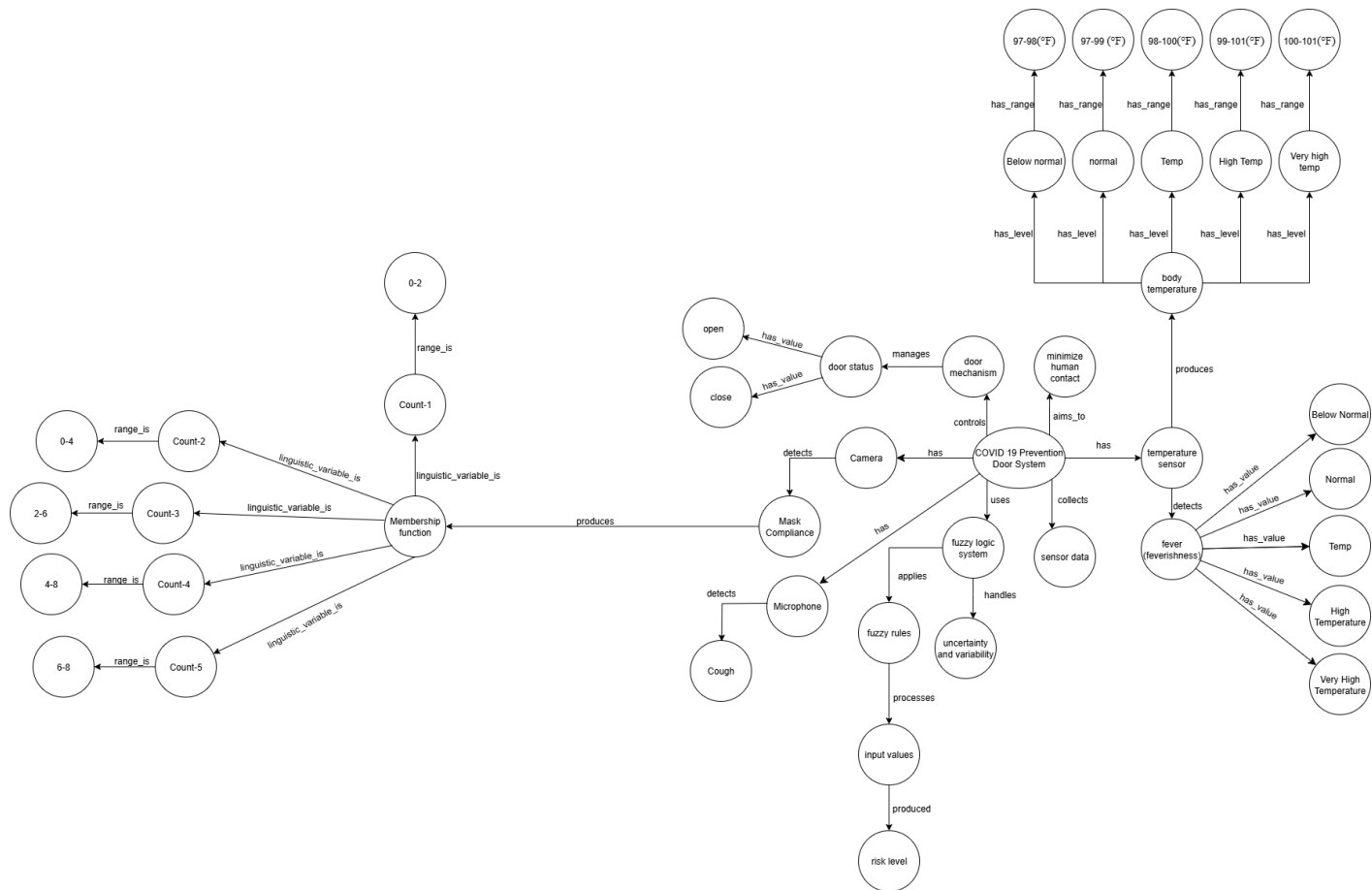


Figure 1: Semantic network of Covid 19 Prevention Door System Fuzzy



3.2(B) FUZZY REPRESENTATION TABLE, FUZZY LINGUISTIC VARIABLES AND VALUES

Fuzzy Representation Table

Body Temperature

Membership Function	Ranges (°F)
Below Normal	97-98
Normal	97-99
Temp	98-100
High Temperature	99-101
Very High Temperature	100-101

Table 1: Body Temperature Fuzzy Representation Table

Mask Compliance

Membership Function	Ranges
Count-1	0-2
Count-2	0-4
Count-3	2-6
Count-4	4-8
Count-5	6-8

Table 2: Mask Compliance Fuzzy Representation Table

Fuzzy Linguistic Variables and Values

1. **Body Temperature:** Below Normal, Normal, Temp, High Temperature, Very High Temperature
2. **Mask Compliance:** Count-1, Count-2, Count-3, Count-4, Count-5
3. **Door Status:** Open, Close



3.2(C) GRAPH REPRESENTATION

The graph representation shows how input like Body temperature and Mask compliance are fuzzified and processed through a rule base.

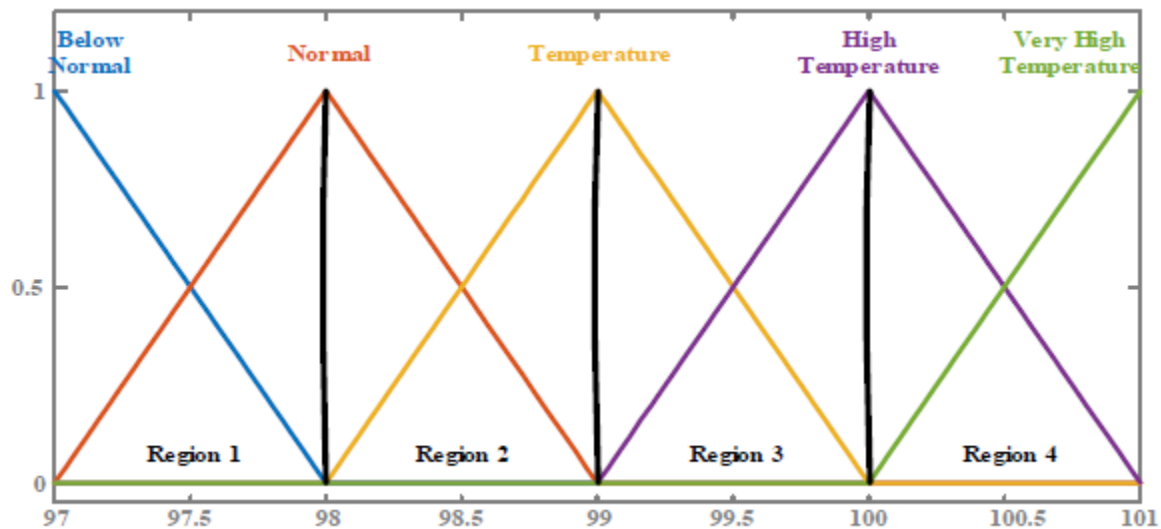


Figure 2: The graph shown the input variable for "Body temperature"

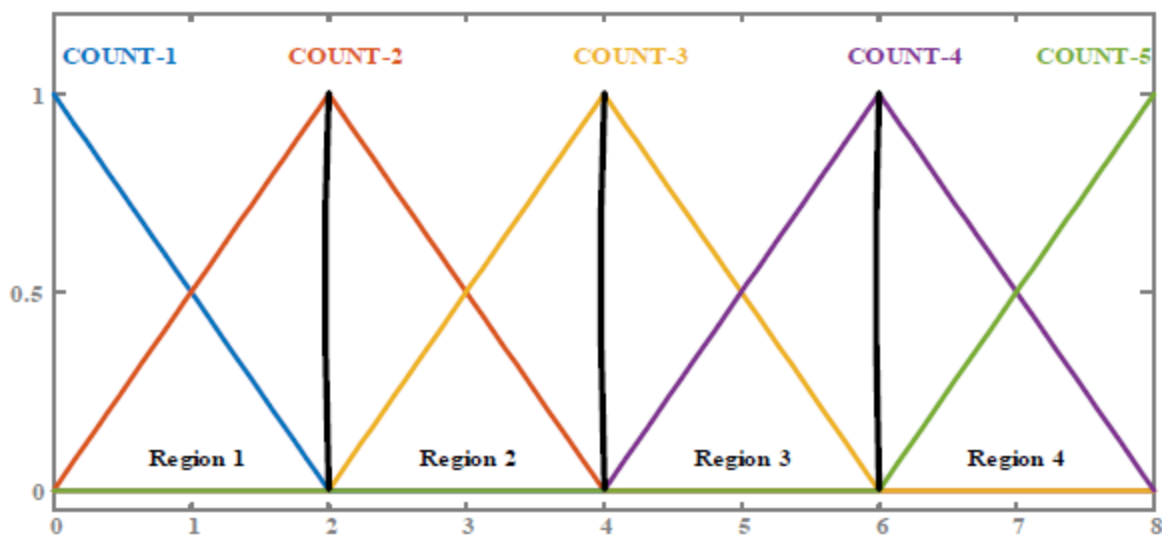


Figure 3: The graph shown the input variable for "Mask Compliance"



3.2(D) FUZZY RULES

Rule 1

IF Mask Compliance = Count-1
OR Body Temperature = Below Normal
THEN Door = Open

Rule 2

IF Mask Compliance = Count-1
OR Body Temperature = Normal
THEN Door = Open

Rule 3

IF Mask Compliance = Count-1
AND Body Temperature = Temp
THEN Door = Close

Rule 4

IF Mask Compliance = Count-1
AND Body Temperature = High Temperature
THEN Door = Close

Rule 5

IF Mask Compliance = Count-1
AND Body Temperature = Very High Temperature
THEN Door = Close

Rule 6

IF Mask Compliance = Count-2
OR Body Temperature = Below Normal
THEN Door = Open

Rule 7

IF Mask Compliance = Count-2
OR Body Temperature = Normal
THEN Door = Open

**Rule 8**

IF Mask Compliance = Count-2
AND Body Temperature = Temp
THEN Door = Close

Rule 9

IF Mask Compliance = Count-2
AND Body Temperature = High Temperature
THEN Door = Close

Rule 10

IF Mask Compliance = Count-2
AND Body Temperature = Very High Temperature
THEN Door = Close

Rule 11

IF Mask Compliance = Count-3
AND Body Temperature = Below Normal
THEN Door = Close

Rule 12

IF Mask Compliance = Count-3
AND Body Temperature = Normal
THEN Door = Close

Rule 13

IF Mask Compliance = Count-3
OR Body Temperature = Temp
THEN Door = Close

Rule 14

IF Mask Compliance = Count-3
AND Body Temperature = High Temperature
THEN Door = Close

Rule 15

IF Mask Compliance = Count-3
AND Body Temperature = Very High Temperature
THEN Door = Close

**Rule 16**

IF Mask Compliance = Count-4
AND Body Temperature = Below Normal
THEN Door = Close

Rule 17

IF Mask Compliance = Count-4
AND Body Temperature = Normal
THEN Door = Close

Rule 18

IF Mask Compliance = Count-4
OR Body Temperature = Temp
THEN Door = Close

Rule 19

IF Mask Compliance = Count-4
OR Body Temperature = High Temperature
THEN Door = Close

Rule 20

IF Mask Compliance = Count-4
AND Body Temperature = Very High Temperature
THEN Door = Close

Rule 21

IF Mask Compliance = Count-5
AND Body Temperature = Below Normal
THEN Door = Close

Rule 22

IF Mask Compliance = Count-5
AND Body Temperature = Normal
THEN Door = Close

**Rule 23**

IF Mask Compliance = Count-5

OR Body Temperature = Temp

THEN Door = Close

Rule 24

IF Mask Compliance = Count-5

OR Body Temperature = High Temperature

THEN Door = Close

Rule 25

IF Mask Compliance = Count-5

AND Body Temperature = Very High Temperature

THEN Door = Close

3.3 SAMPLE INPUT

To study the fuzzy inference system for our door with the fuzzy sets and rules that we have gathered, here are 3 samples of input to show the ways our system produces output, whether the door status is open or closed.

Sample Input 1:

Body Temperature: 97.5

Mask Compliance: 1.5

Output: 0.970 (Door is Open)

Sample Input 2:

Body Temperature: 98.7

Mask Compliance: 4.1

Output: 0.340 (Door is Close)

Sample Input 3:

Body Temperature: 99.6

Mask Compliance: 7.5

Output: 0.030 (Door is Close)



3.4 INFERENCE SYSTEM

The fuzzy Inference System (FIS) is a method that uses fuzzy sets to map inputs to other outputs Covid 19 Prevention Door System design implements the Mamdani inference approach. Mamdani's fuzzy inference process has a 4 step procedure of input variables which is fuzzification, rule evaluation, aggregation, and defuzzification. To construct the Mamdani fuzzy model, the system requires inferencing results from both input and output. The Mamdani system is more suitable and preferable than Sugeno style because of its simple rules bases and easy to understand rules which are generated by human experts. An IF-THEN statement is used to represent these rules that are designed using if-then statements. Once the input variables have been fuzzified, a fuzzy rule will be applied by the Inference System.

3.4(A) FUZZIFICATION

Fuzzification refers to the process of the crisp input to fuzzy value. We also define the membership functions of body temperature and mask compliance using the formula of a straight line which is $y = mx + c$.

Sample Input 1

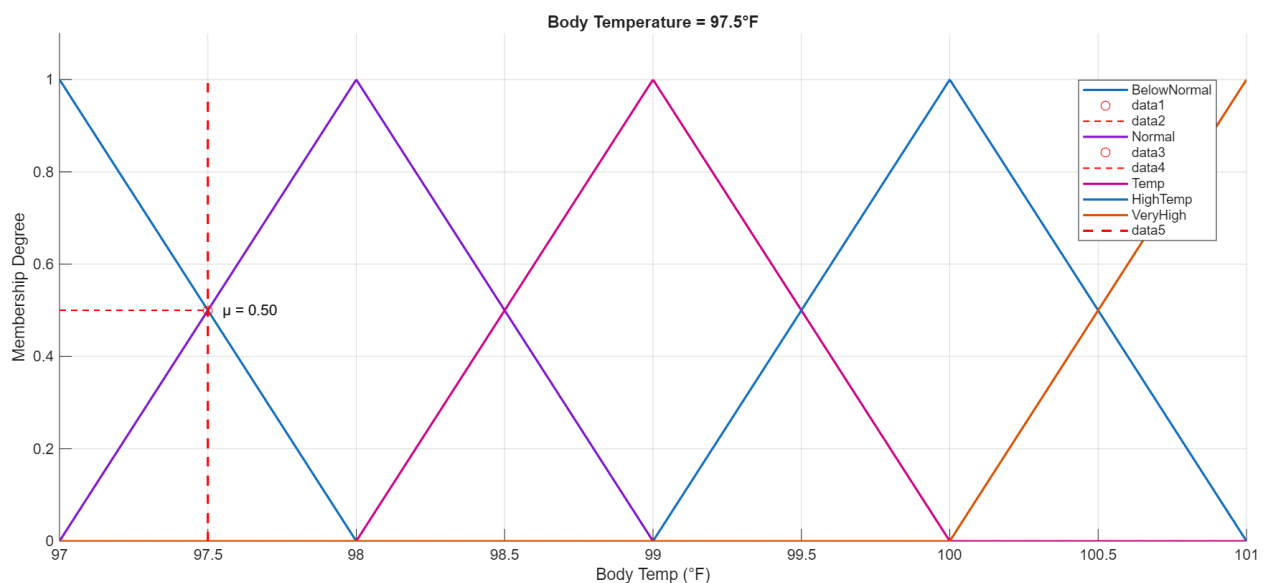


Figure 4: Input 1 graph (Body Temperature: 97.5)

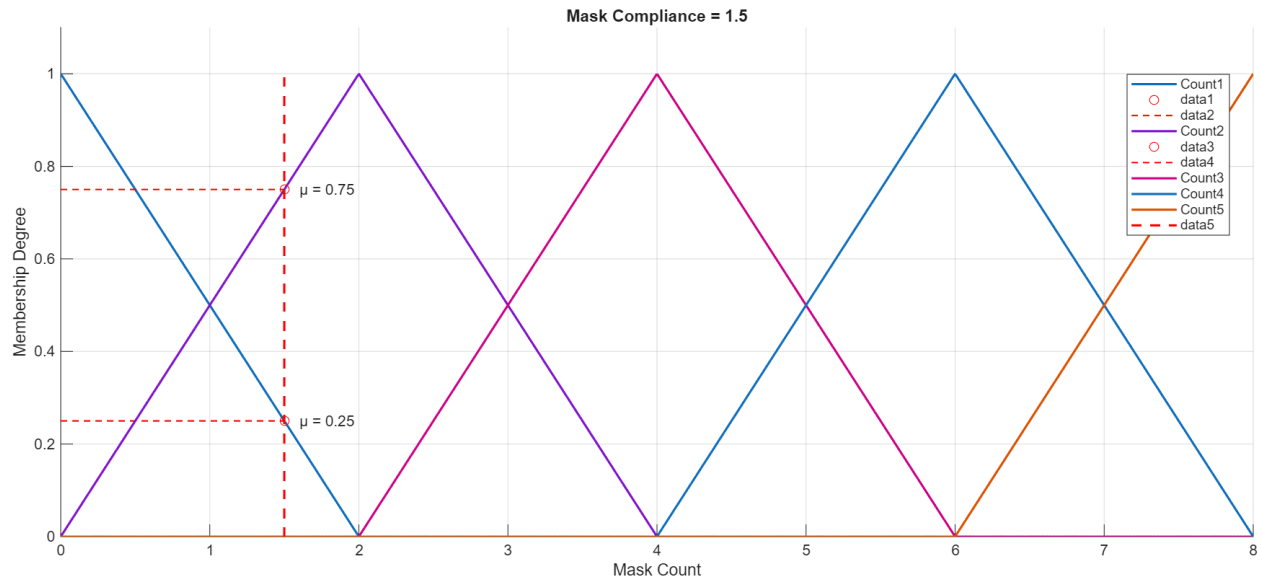


Figure 5: Input 2 graph (Mask Compliance= 1.5)

Sample Input 2

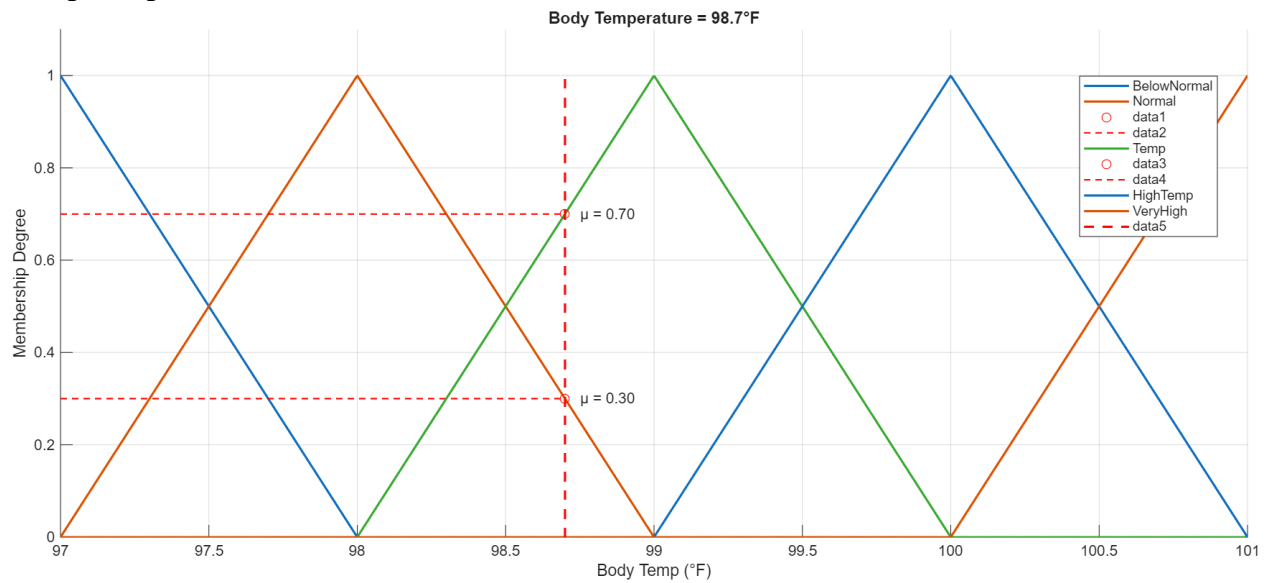


Figure 6: Input 1 graph (Body Temperature: 98.7)

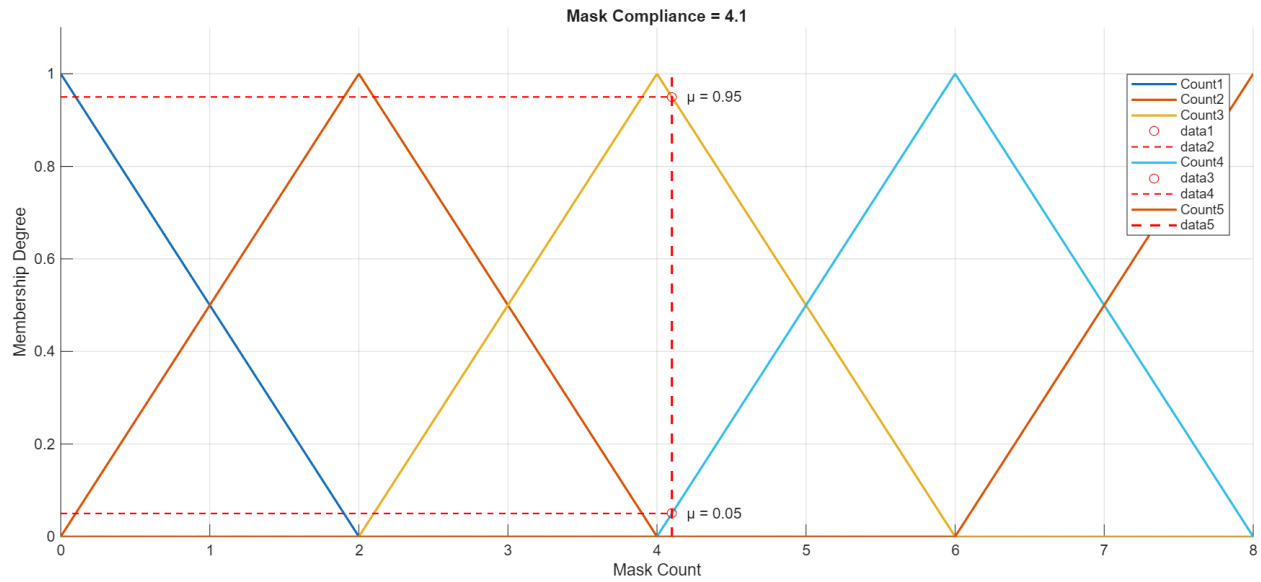


Figure 7: Input 2 graph (Mask Compliance: 4.1)

Sample Input 3

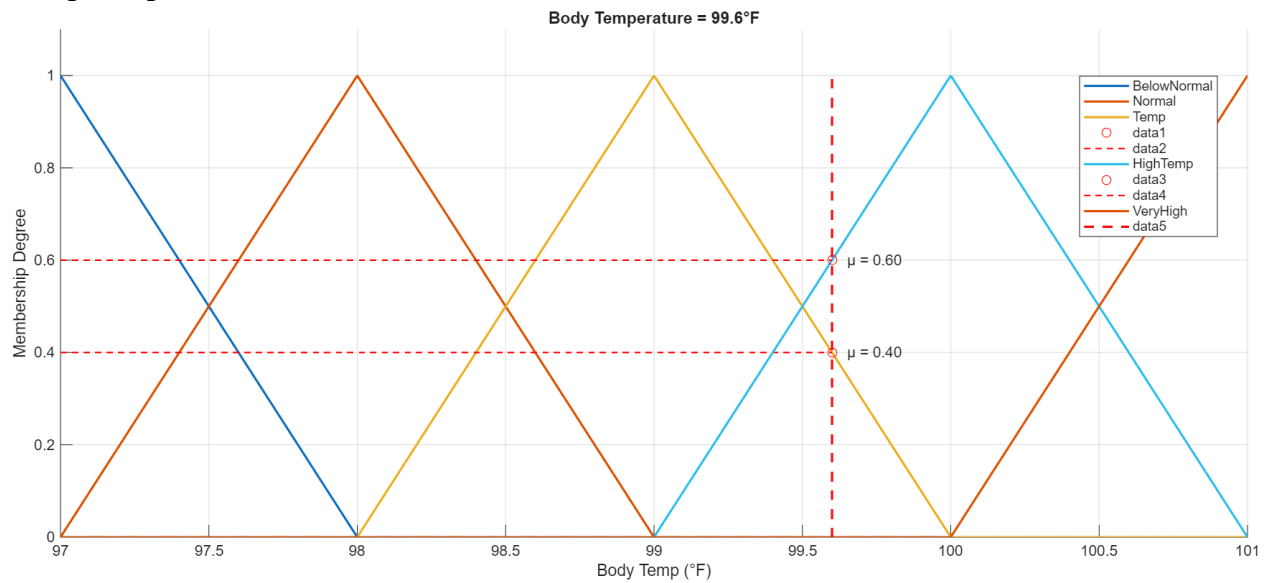


Figure 8: Input 1 graph(Body Temperature: 99.6)

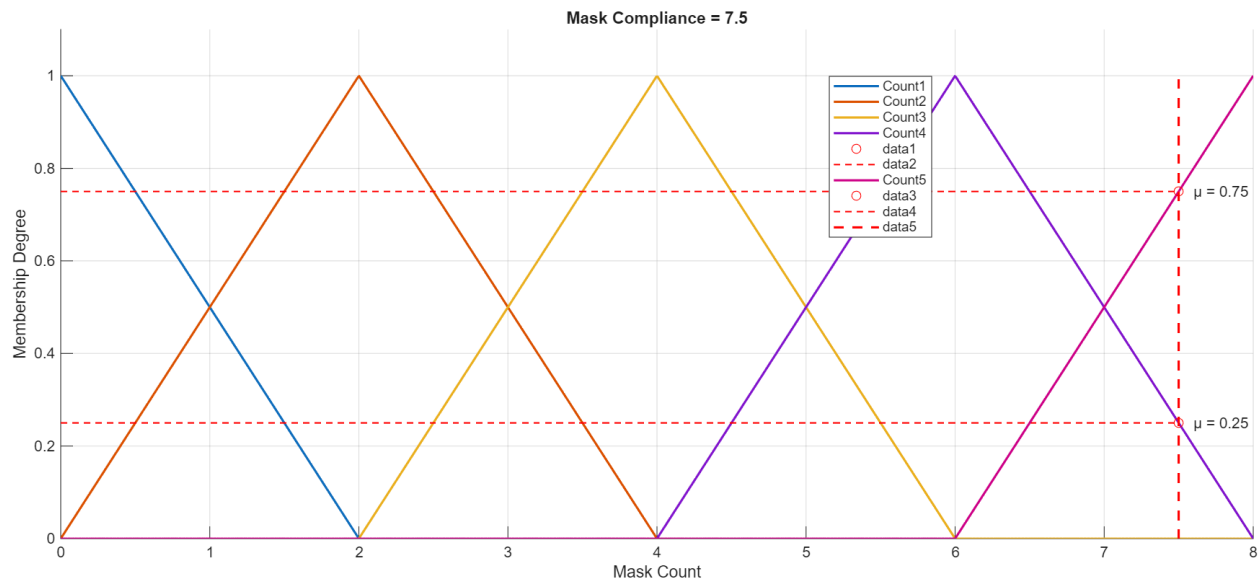


Figure 9: Input 2 graph(Mask Compliance: 7.5)

3.4(B) RULE EVALUATION

Rule evaluation is a process that combines the antecedent that applies the fuzzy operator (AND or OR) to get a single number that represents the outcome of the antecedent evaluation. In this process, we are using Mamdani-style inference to evaluate this.

AND fuzzy operation intersection:

$$\mu A(x) \text{ AND } \mu B(x) \equiv \min[\mu A(x), \mu B(x)]$$

OR fuzzy operation union:

$$\mu A(x) \text{ OR } \mu B(x) \equiv \max[\mu A(x), \mu B(x)]$$

Sample Input 1

Rule	Rule Evaluation	Output
Rule 1	IF Mask Compliance = Count-1 {0.25} OR Body Temperature = Below Normal {0.50} THEN Door = Open	$\max(0.25, 0.50) = 0.50$



Rule 2	IF Mask Compliance = Count-1 {0.25} OR Body Temperature = Normal {0.50} THEN Door = Open	$\max(0.25, 0.50) = 0.50$
Rule 6	IF Mask Compliance = Count-2 {0.75} OR Body Temperature = Below Normal {0.50} THEN Door = Open	$\max(0.75, 0.50) = 0.75$
Rule 7	IF Mask Compliance = Count-2 {0.75} OR Body Temperature = Normal {0.50} THEN Door = Open	$\max(0.75, 0.50) = 0.75$

Table 3: Sample 1 Rule Evaluation

Sample Input 2

Rule	Rule Evaluation	Output
Rule 12	IF Mask Compliance = Count-3 {0.95} AND Body Temperature = Normal {0.30} THEN Door = Close	$\min(0.95, 0.30) = 0.30$
Rule 13	IF Mask Compliance = Count-3 {0.95} OR Body Temperature = Temp {0.70} THEN Door = Close	$\max(0.95, 0.70) = 0.95$
Rule 17	IF Mask Compliance = Count-4 {0.05} AND Body Temperature = Normal {0.30} THEN Door = Close	$\min(0.05, 0.30) = 0.05$
Rule 18	IF Mask Compliance = Count-4 {0.05} OR Body Temperature = Temp {0.70} THEN Door = Close	$\max(0.05, 0.70) = 0.70$

Table 4: Sample 2 Rule Evaluation

Sample Input 3

Rule	Rule Evaluation	Output
Rule 18	IF Mask Compliance = Count-4 {0.25} OR Body Temperature = Temp {0.40} THEN Door = Close	$\max(0.25, 0.40) = 0.40$



Rule 19	IF Mask Compliance = Count-4 {0.25} OR Body Temperature = High Temperature {0.60} THEN Door = Close	$\max(0.25, 0.60) = 0.60$
Rule 23	IF Mask Compliance = Count-5 {0.75} OR Body Temperature = Temp {0.40} THEN Door = Close	$\max(0.75, 0.40) = 0.75$
Rule 24	IF Mask Compliance = Count-5 {0.75} OR Body Temperature = High Temperature {0.60} THEN Door = Close	$\max(0.75, 0.60) = 0.75$

Table 5: Sample 3 Rule Evaluation

3.4(C) AGGREGATION

Aggregation refers to the procedure of combining the fuzzy sets which describe the output of every rule added to yield one fuzzy set. Only one aggregation is done per output variable, which is before the final defuzzification step. The input of the aggregation process is the list of truncated output functions returned by the implication process for each rule. The result of this process is a single combined fuzzy set corresponding to each output variable.

Sample Input 1

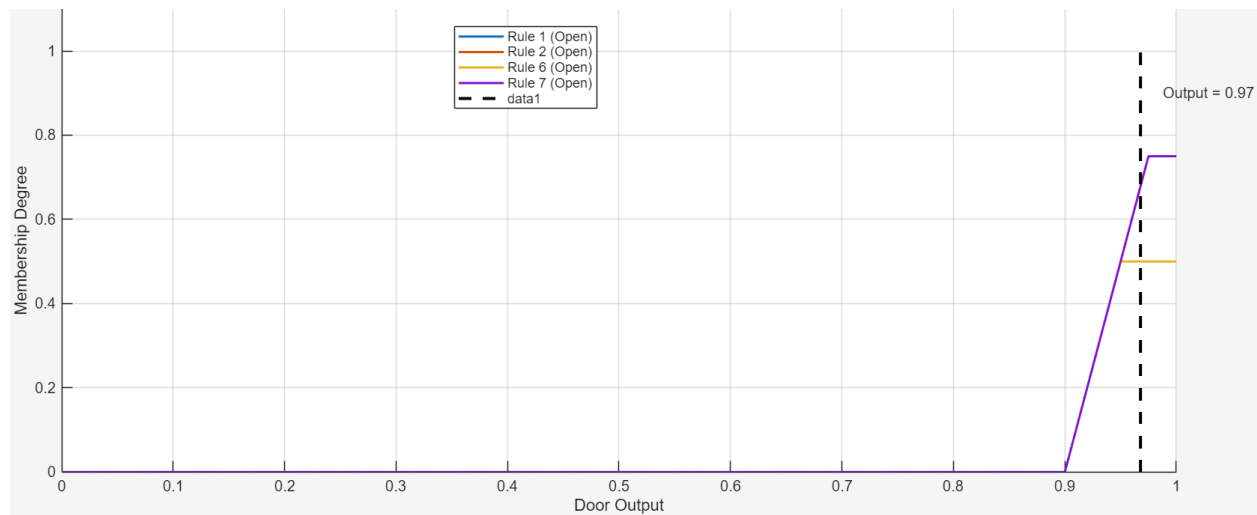


Figure 10: Aggregation Sample Input 1



Sample Input 2

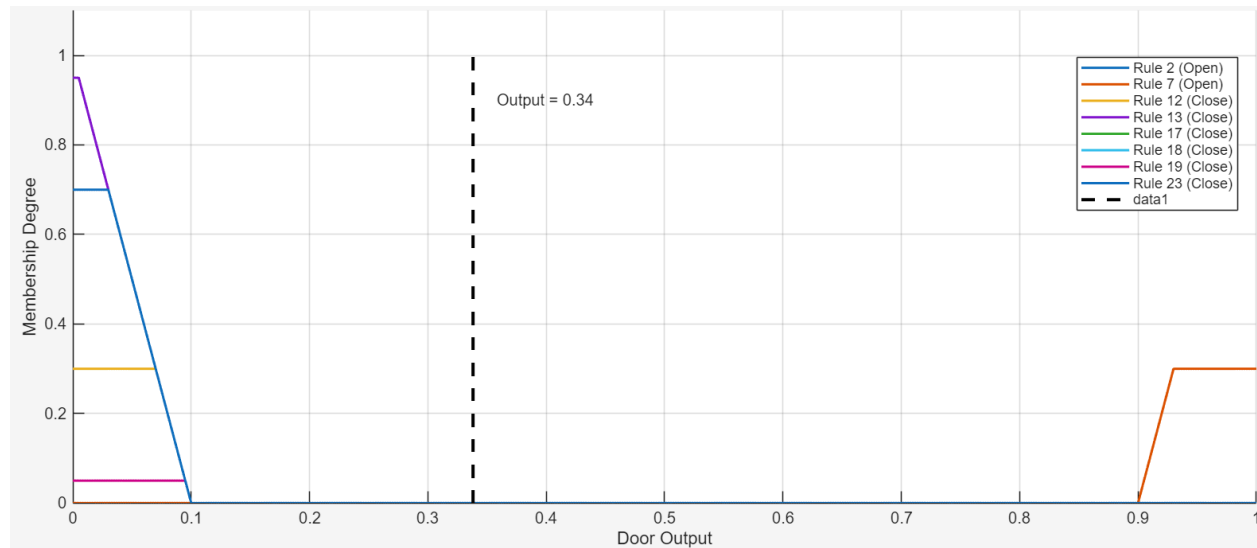


Figure 11: Aggregation Sample Input 2

Sample Input 3

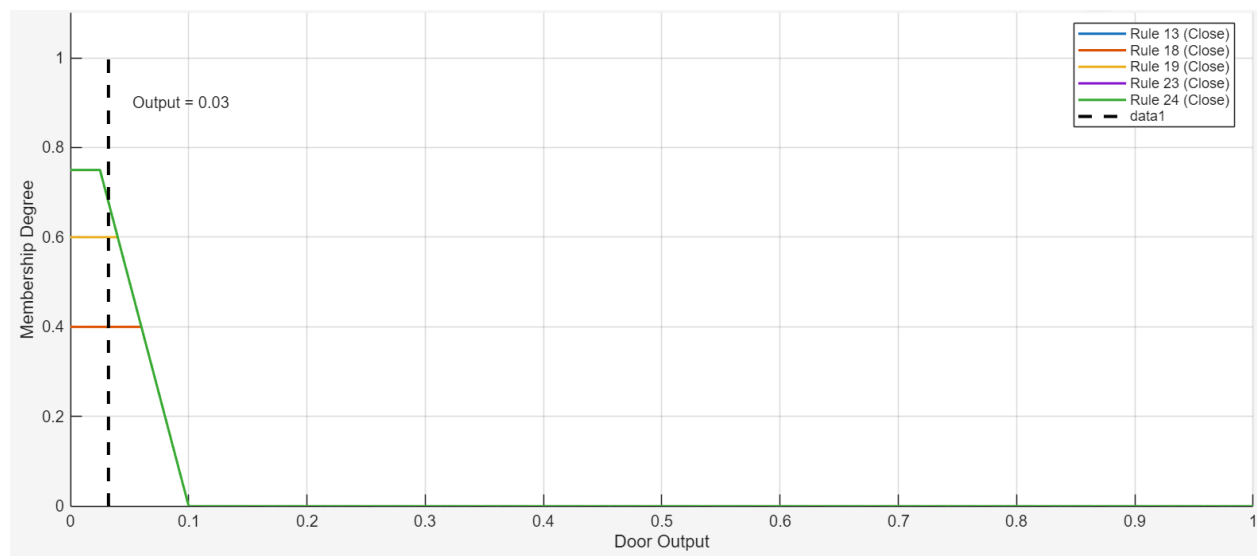


Figure 12: Aggregation Sample Input 3



3.4(D) DEFUZZIFICATION

Defuzzification is a process of turning a fuzzy set into a single number value. Defuzzification procedures are classified into two types: bisector and centroid. The centroid approach is the most often used defuzzification method, as it returns the center of the region under the curve. Center of Gravity (COG) = $\sum (\mu_i \times x_i) / \sum \mu_i$

Sample Input	Defuzzification (Centroid Technique)
Sample 1 Body Temperature: 97.5°F Mask Compliance: 1.5	$\text{COG} = \frac{(0.98 \times 0.75) + (0.99 \times 0.75)}{(0.75 + 0.75)}$ $= \frac{1.478}{1.500}$ $= 0.985$ $= 98.5\%$
Conclusion Sample 1	The calculated door status, based on the defuzzification result of 97% for "Open", when the input is a Body Temperature (BT) of 97.5°F and Mask Compliance (MC) of 1.5. This indicates that the door is most likely to open, as the body temperature is considered normal and the mask compliance is within an acceptable range. Hence, the door is 98.5% likely to open, indicating normal health conditions.
Sample 2 Body Temperature: 98.7°F Mask Compliance: 4.1	$\text{COG} = 0.340$ $= 34\%$
Conclusion Sample 2	The calculated door status, based on the defuzzification result of 34% for "Close", when the input is a Body Temperature (BT) of 98.7°F and Mask Compliance (MC) of 4.1. Since the aggregation of fuzzy set is "Close", it is too small to be calculated manually. Thus, it consider the result from MATLAB modelling that the output is 34%. This suggests that the door is more likely to close, as the temperature is borderline, and the mask compliance is moderate, triggering a safety precaution to prevent access. Hence, the door is 34% likely to open, meaning moderate risk based on the combination of inputs.
Sample 3 Body Temperature: 99.6°F Mask Compliance: 7.5	$\text{COG} = \frac{(0.01 \times 0.75) + (0.02 \times 0.75)}{(0.75 + 0.75)}$ $= \frac{0.023}{1.500}$ $= 0.015$ $= 1.5\%$



Conclusion Sample 3

The calculated door status, based on the defuzzification result of 1.5% for "Close", when the input is a Body Temperature (BT) of 99.6°F and Mask Compliance (MC) of 7.5. This indicates that the door is highly likely to remain closed, as both the body temperature is considered high and the mask compliance is low, which poses a higher health risk, thus preventing entry. Hence, the door is 1.5% likely to open, with high risk due to both elevated temperature and poor mask compliance.

Table 6: Defuzzification for all Sample Input

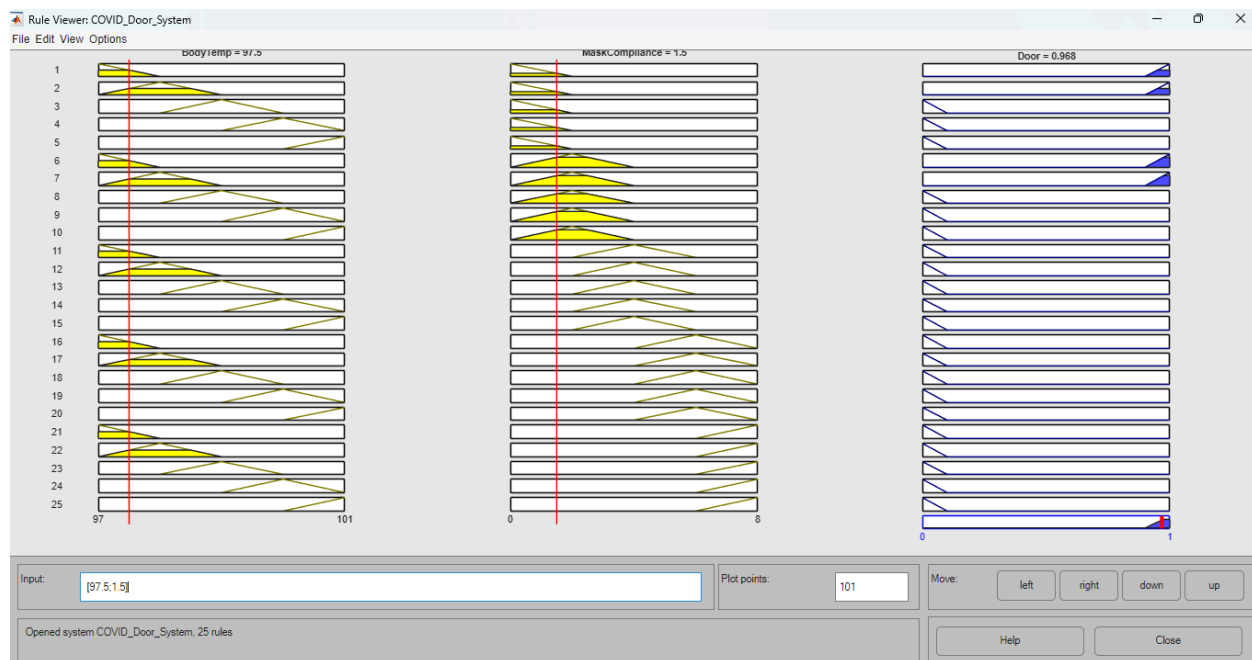


Figure 13: Defuzzification for sample input 1

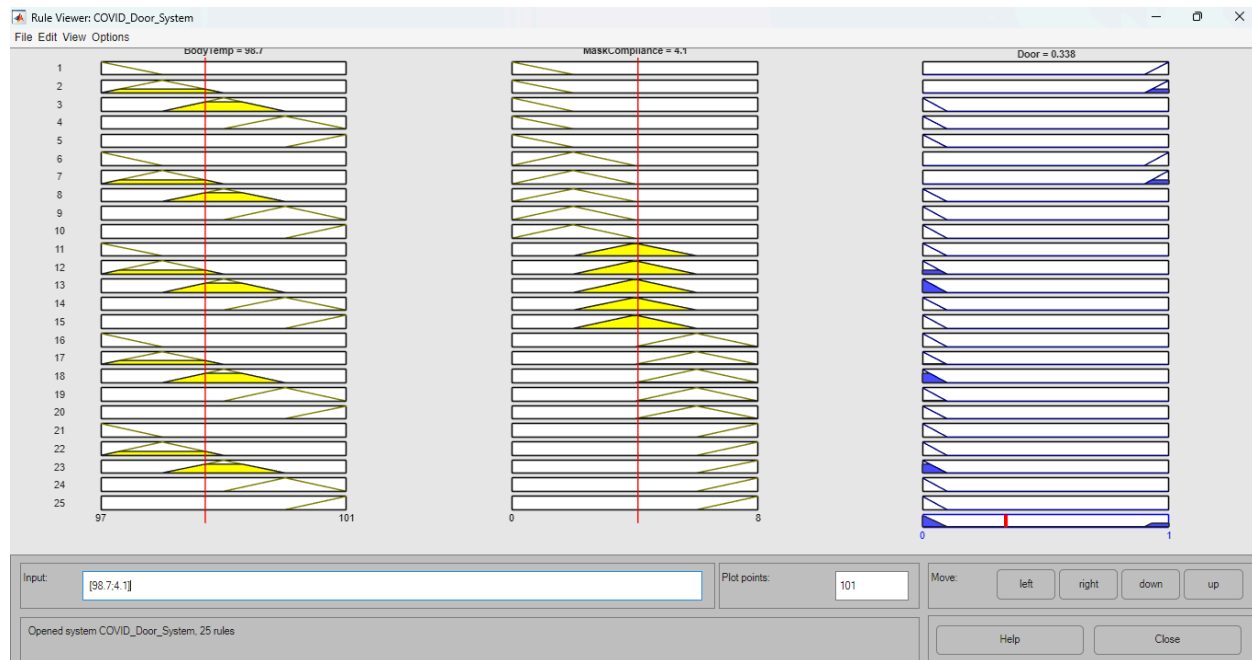


Figure 14: Defuzzification for sample input 2

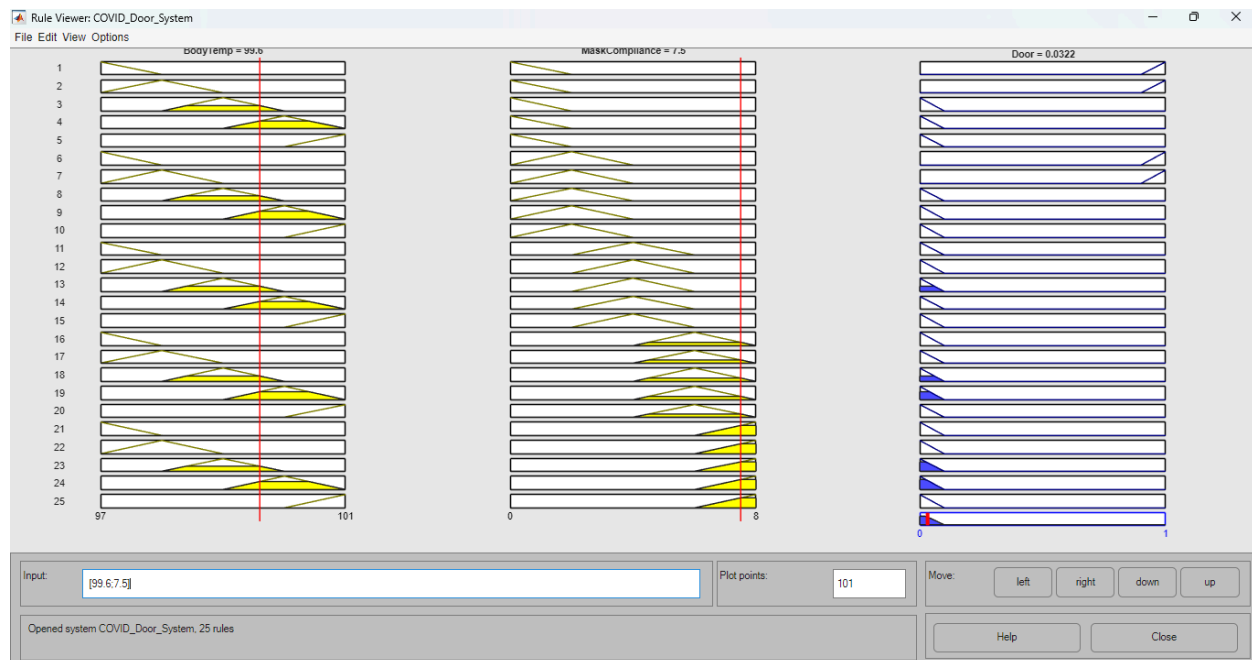


Figure 15: Defuzzification for sample input 3



4.0 RESULT AND ANALYSIS

The testing of fuzzy logic-based COVID-19 prevention door systems involved three sample inputs, combining body temperature and mask compliance. The entire inputs were processed through Mamdani fuzzy inference and centroid defuzzification providing output values in form of the center of gravity (COG) determining whether to have the door open or closed. Here the results and analysis for each samples:

1. Sample 1

Input;

Body Temperature: 97.5°F

Mask Compliance: 1.5

Output: 0.985(Door is Open)

In this case, the body temperature and the mask compliance is safe. The temperature is categorized as “Below Normal” and “Normal” and the mask compliance value of 1.5 falls between count-1 and count-2 in the fuzzy set. Based on fuzzy rules, such as Rule 1 and Rule 2, these inputs produce a high degree of membership in the output of the “Open” considering the fuzzy rules. With the centroid defuzzification, this value turns out to be 0.985 and is beyond the threshold which means that the door will open. It means that the health risk is low and the system operates correctly.

2. Sample 2

Input;

Body Temperature: 98.7°F

Mask Compliance: 4.1

Output: 0.340 (Door is Close)



This input represents a borderline case. The body temperature is in the zone between “Normal” and “Temp” while the mask compliance value of 4.1 reflects the higher count within the fuzzy range count-3 to count-4. According to fuzzy rules like Rule 13 and Rule 18 which deals with medium mask adherence cases and those involving increase in temperature, the system appropriately comes up with this as more dangerous. The center of gravity(COG) is less than the threshold of opening and the decision reached is to keep the door closed. This is an indication of the capacity of the system to implement fuzzed reasoning in processing of overlapping and uncertain inputs in the name of implementing safety.

3. Sample 3

Input;

Body Temperature: 99.6°F

Mask Compliance: 7.5

Output: 0.015 (Door is Close)

The situation under this case involves both high body temperature and low mask compliance which could mean health risk. The mask compliance is 7.5 that is within the range of count-5 or least compliant group while the temperature is identified as “High Temperature”. The fuzzy rules including Rule18, Rule 19, Rule 23 and Rule 24 have a strong connection with rules and this combination with such fuzzy rules are strongly related to a decision if the door is closed. The last defuzzified value is 0.015, which clearly shows that the door has to remain closed. This confirms that the system is responsive to critical inputs that indicate risk of infection or lack of compliance.



The benefits can be realized through its outcome indicating that the fuzzy system of logic is effective in determining whether an entry is safe or dangerous. When the Mamdani and centroid defuzzification rule is applied, the decision is logical and interpretable, and it fits the expectation of real world application. Fuzzy logic had the flexibility to deal with uncertainties that might be minor temperature changes, imprecise mask detection and still perform consistently. The rule system also could resemble human-like decision making and assess the set of inputs that cannot be accurately defined as safe or unsafe but are considered with some level of confidence. This demonstrates the fact that fuzzy logic offers a less abstraction and more flexible framework relative to systems of binary decisions.



5.0 CONCLUSION

The implementation of the COVID-19 Prevention Door System based on fuzzy logic has been beneficial to society aiming to improve health safety with contactless screening. By combining the two features, mask detection, and non-contact temperature measurement, along with the use of a fuzzy logic decision-making system, the system works to achieve a satisfactory result in determining whether or not a person can enter a facility based on their safety level. The Mamdani inference system and the centroid defuzzification allowed the system to deliver accurate and interpretable results to different levels of compliance and body temperature.

The three sample scenario testing also revealed that the use of fuzzy logic provides a greater flexibility in managing the uncertainties in the real world that cannot be easily managed in binary logic systems given its ability to care for borderline temperatures and intra-mask behavior. The output of the system corresponded to the anticipated health risk profiling giving the assurance that only those with healthy health indicators will have access

Comprehensively, this project substantiates fuzzy logic methodology in implementing intelligent and automated access control systems to the outbreak of a public health crisis. It demonstrated opportunities to develop further in other health related screening activities and is flexible to meet evolving conditions and physiological standards, and therefore is scalable and marketable to expand scope of its application beyond the COVID-19 pandemic.



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